Quinn Gifford

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EDUCATION

Oregon State University — **Corvallis, OR**

B.S. in Electrical and Computer Engineering, Computer Science Minor | GPA: 3.90 — **June 2027 (Expected)**

TECHNICAL SKILLS

Embedded Systems: Linux, C, C++, Firmware, PCB Design, KiCad, Soldering, Debugging, RTOS, STM32, I2C, UART, SPI
Digital Design & Architecture: SystemVerilog, FPGA, Cadence, Quartus, RISC-V, AVR, Chisel, Microchip Studio, VLSI, CMOS
Electrical Analysis: LTSpice, Matlab, Simulink, Modelsim, Digital Signal Processing, Power Systems, Verification & Validation
Software Engineering: Python, C#, Java, JavaScript, TypeScript, React, Node.js, Next.js, Git, CI/CD, Docker, PostgreSQL, SQL, Supabase, AWS, Vercel, Render, Pinecone, Stripe

PROFESSIONAL EXPERIENCE

Systems Engineer Intern | Boeing — **June 2026 - September 2026**

- Contributed to verification, validation, and certification of the Earth Reference System and Air Data and Inertial Reference System.
- Used Matlab data plotting, Simulink model testing, and lab simulations to verify system requirements using real flight data and create official documentation to show compliance with FAA regulations.
- Used engineering judgement to identify, track, and resolve requirement deviations. Collaborated with team and suppliers to coordinate version control of documents and spreadsheets pertaining to deviations.

Software Engineer Intern | Alethian — **June 2025 - September 2025**

- Deployed a national-scale vector embedding database using Pinecone to store and index pharmacy data enriched with geolocation metadata and entry-type filters, enabling highly performant semantic queries.
- Built a high-speed pharmacy search API endpoint using FastAPI, combining semantic vector search with geospatial proximity ranking; achieved **sub-100ms latency** for targeted queries.
- Created performance testing tools for various applications such as insurance card scanning and speech recognition, allowing our team to make up to **18% increases in model accuracy** through iterative improvements.

Software Engineer (Contract) | Scale — **May 2024 - September 2024**

- Pre-processed GitHub pull requests from SWE-bench subset for LLM coding training using GitHub API and Pandas, resulting in a **12% increase** in resolved programming problems during testing.
- Completed RLHF tasks, including writing programming related prompts, evaluating model generated code, and writing correct code for model training, in various languages such as Python, Java, JavaScript, Go, PHP, Verilog.

STEM Instructor | STEP IT Academy — **June 2023 - September 2023**

- Instructed classes of up to 10 students on various technology related subjects such as robotics, digital logic design, data structures and algorithms, game development, Minecraft modding, and more.
- Assisted with technology integration in the classrooms, troubleshooting technical issues, and maintaining digital learning resources.

PROJECTS (see website for more →) — https://quinngiffordwebsite.vercel.app/

Model Rocket Flight Computer

- Designed and built a STM32F411-based flight data recorder PCB with RTOS firmware, power management, real-time sensor logging, and robust onboard data capture for model rocket flight testing and post-flight analysis.
- Integrated an IMU and TMP117 temperature sensor over I2C and logged flight data to a microSD card over SPI using an RTOS-based task structure for scheduled sampling, buffering, storage, fault handling, and reliable operation under flight conditions.
- Developed firmware for sensor acquisition, timestamping, data buffering, and reliable flash storage, and created a post-flight Python tool to convert logs to CSV, generate plots, and support detailed analysis, debugging, and performance review.
- Performed PCB bring-up, and low-level debugging of I2C, SPI, and GPIO card interfaces using an oscilloscope and circuit debugger.

Guitar Hero on FPGA

- Recreated Guitar Hero using a CMOD S7 FPGA to handle game logic and drive a four-strip LED strip display, along with a custom ESP32-based PCB that serves as the wireless game controller.
- Wrote SystemVerilog code on the FPGA to control timing-accurate note generation, input processing, score keeping, and LED-based gameplay visualization, along with C++ firmware on the ESP32 to handle button inputs and bluetooth transmission to the FPGA.
- Created a beatmap generator in Python that converts song audio into an array of sub-millisecond-accurate beat timings, using digital signal processing (DSP) to detect the timing, strength, and instrument classification of each beat. The generator creates multiple difficulty levels and runs in ~200 milliseconds. Programmed the FPGA to receive, store, and manage these beatmaps for level selection.